LLMs in Drug Research: Advancements, Methods, Challenges, and Future Directions

Abstract

This review synthesizes the current landscape of Large Language Models (LLMs) in drug research, highlighting their potential to revolutionize pharmaceutical methodologies. It explores key themes such as automation of drug discovery, knowledge extraction, and explainability. Methods include specialized and general-purpose LLMs, multi-agent systems exemplified by PharmAgents, and computational tools for drug candidate evaluation. The review identifies challenges including data scarcity, bias, methodological limitations, computational constraints, and ethical considerations like data privacy and interpretability. It emphasizes the need for robust data governance, standardized evaluation metrics, and transparent systems. A proposed research pipeline is presented, focusing on data curation, target identification, lead optimization, preclinical evaluation, and clinical trial design, alongside explainability analysis. Future research should focus on improved data sets, validation methods, and LLM transparency to fully leverage their potential for accelerating drug discovery and improving patient outcomes. LLMs show promise, but addressing these challenges is crucial for responsible and impactful innovation in this rapidly evolving field.

Introduction

The intersection of Large Language Models (LLMs) and drug research represents a rapidly evolving frontier with the potential to revolutionize traditional pharmaceutical methodologies. Drug discovery and development, historically characterized by high costs, lengthy timelines, and significant attrition rates, stand to benefit immensely from the integration of LLMs. LLMs offer the promise of accelerating various stages of the drug development pipeline, from target identification and lead optimization to preclinical evaluation and clinical trial design [2, 1]. This review addresses the novel methods, evaluation metrics, and ethical considerations that underpin the application of LLMs in drug research. The current landscape includes specialized LLMs trained on scientific languages and general-purpose LLMs adapted for biomedical tasks [2].

This review synthesizes insights from recent studies and implementations, including analyses of virtual pharmaceutical ecosystems powered by LLM-driven multi-agent collaboration [1]. The goal is to provide a critical overview of the current state-of-the-art, highlighting both the advancements and the limitations of LLMs in this domain. Specifically, this review will delve into how LLMs are being employed to understand disease mechanisms, predict drug efficacy and safety, and streamline clinical trials [2]. It will also explore the challenges and opportunities associated with these applications, focusing on issues such as data privacy, bias mitigation, and the need for robust evaluation metrics.

The significance of this research area lies in its potential to transform drug discovery from a largely empirical and reactive process to a more rational, predictive, and proactive endeavor. LLMs can process vast amounts of biomedical data, identify patterns and relationships that might be missed by human researchers, and generate novel hypotheses for drug targets and therapeutic interventions. However, the successful integration of LLMs into drug research requires careful consideration of several factors, including the quality and representativeness of the training data, the interpretability and explainability of the models, and the ethical implications of their use. For instance, the PharmAgents framework exemplifies an LLM-based multi-agent system designed to simulate the entire drug discovery workflow, showcasing the potential for autonomous and scalable pharmaceutical research [1]. Yet, the complexity and scale of such systems necessitate rigorous evaluation and validation to ensure their reliability and trustworthiness. This review aims to contribute to a deeper understanding of these issues and to provide guidance for future research and development in this rapidly evolving field. By examining novel methodologies, evaluation practices, and ethical considerations, this review seeks to foster responsible and impactful innovation at the intersection of artificial intelligence and pharmaceutical science. The ultimate aim is to accelerate the discovery and development of new and effective treatments for diseases, thereby improving human health and well-being. Furthermore, exploring the integration of specialized machine learning models and computational tools within LLM frameworks, as seen in PharmAgents, can improve decision-making across the

drug discovery pipeline [1]. The discussion will encompass the current usage of LLMs, future research possibilities, and the inherent challenges in this rapidly growing sector.

Key Themes and Trends

Several key themes and trends emerge from the literature regarding the application of LLMs in drug research. A primary theme is the **automation of drug discovery workflows** through LLM-driven agent systems [1]. This involves leveraging LLMs to simulate and streamline various stages of the drug development pipeline, from target identification to preclinical evaluation. PharmAgents, as described in [1], exemplifies this trend by creating a virtual pharmaceutical ecosystem where specialized LLM agents collaborate to mimic the real-world drug discovery process. Each agent is equipped with domain-specific tools and models, facilitating rational, transparent, and high-performance decision-making across the entire pipeline. This approach contrasts sharply with traditional methods that are often complex, resource-intensive, and time-consuming, highlighting the potential of LLMs to accelerate and optimize the drug discovery process.

Another prominent trend is the **utilization of LLMs for knowledge extraction and integration** from diverse biomedical data sources [2]. LLMs can effectively analyze and interpret vast amounts of unstructured and structured data, including scientific literature, chemical databases, and clinical trial reports, to identify potential drug targets, predict drug efficacy and safety, and understand disease mechanisms. As mentioned in [1], agents access databases like the Drug Target Database and UniProt to identify therapeutic targets. This capability is particularly valuable for enhancing target-disease linkage and generating novel hypotheses in drug discovery. The review paper [2] categorizes LLMs into specialized LLMs trained on specific scientific languages and general-purpose LLMs trained on diverse textual data, emphasizing the versatility of LLMs in handling various types of biomedical information.

Explainability and interpretability also represent a significant theme, particularly within the context of LLM-driven drug discovery [1]. Unlike many traditional machine learning models that operate as "black boxes," LLMs have the potential to provide more transparent and interpretable insights into their decision-making processes. For instance, PharmAgents [1] emphasizes the explainability of its agents, allowing researchers to understand the reasoning behind their predictions and recommendations. This is crucial for building trust in LLM-based drug discovery systems and ensuring that their outputs are scientifically sound and clinically relevant. This emphasis on interpretability also facilitates the identification and mitigation of potential biases in training data, which is a critical ethical consideration.

While the automation and knowledge integration capabilities of LLMs are well-established trends, **self-evolution and continuous learning** represent an emerging trend [1]. PharmAgents aims to enable agents to self-evolve to refine future drug designs based on prior experience [1]. This capability allows LLM-based drug discovery systems to continuously improve their performance and adapt to new data and insights. This aspect highlights the potential of LLMs to not only accelerate drug discovery but also to drive innovation and generate novel therapeutic strategies. However, the practical implementation and validation of self-evolution capabilities remain a significant challenge, requiring robust evaluation metrics and careful monitoring to ensure that the system learns effectively and avoids unintended consequences. The success of these trends will depend on addressing ethical considerations such as data privacy, bias, and responsible use.

Methods and Technologies

The application of LLMs in drug research encompasses a diverse range of methods and technologies, reflecting the multifaceted nature of the drug discovery and development pipeline. A prominent approach involves the development of multi-agent systems, exemplified by PharmAgents [1], which seeks to simulate the entire pharmaceutical research process within a virtual environment. PharmAgents employs specialized

LLM-based agents for each stage of drug discovery, including target discovery, lead identification, lead optimization, and preclinical evaluation [1]. Each agent is equipped with domain-specific tools and models, facilitating seamless collaboration and knowledge sharing across the pipeline. This approach contrasts with more modular applications of LLMs, where individual tasks within the drug discovery process are augmented by LLMs, as surveyed in [2].

The review in [2] categorizes LLMs used in drug discovery into two main paradigms: specialized LLMs trained on specific scientific languages (e.g., SMILES, FASTA), and general-purpose LLMs trained on diverse textual data. Specialized LLMs excel at tasks requiring an understanding of chemical structures and biological sequences, such as predicting drug-target interactions or generating novel drug candidates [2]. General-purpose LLMs, on the other hand, are well-suited for tasks involving natural language processing, such as extracting information from scientific literature or generating hypotheses about disease mechanisms [2]. The choice of LLM paradigm depends on the specific task and the availability of training data. PharmAgents [1] leverages both specialized and general-purpose LLMs to integrate domain knowledge, automate key stages, and enable interpretability, agent interaction, and self-evolution.

In the target discovery phase, PharmAgents utilizes a disease expert agent that identifies potential therapeutic targets by accessing Drug Target Databases and UniProt [1]. This agent filters and ranks potential targets based on their association with the user-provided disease description [1]. Similarly, the lead identification phase employs a lead identification agent that utilizes tools such as AutoDock Vina to perform virtual screening of potential drug candidates [1]. The lead optimization phase involves iterative refinement of lead compounds using techniques such as molecular docking and scoring functions, while the preclinical evaluation phase assesses the safety and efficacy of drug candidates using in silico models and simulations [1].

Evaluation metrics for LLMs in drug research are diverse and often task-specific. For tasks involving the generation of novel drug candidates, metrics such as novelty, diversity, and drug-likeness are commonly used [2]. These metrics assess whether the generated compounds are structurally distinct from known drugs, whether the generated set of compounds covers a wide range of chemical space, and whether the generated compounds possess physicochemical properties that make them likely to be orally bioavailable. For tasks involving the prediction of drug-target interactions, metrics such as accuracy, precision, recall, and F1-score are often used [2]. These metrics assess the ability of the LLM to correctly identify true positives, minimize false positives, and minimize false negatives. In the context of PharmAgents [1], evaluation metrics would likely need to assess the overall efficiency and effectiveness of the entire drug discovery pipeline, potentially including metrics such as the time required to identify a promising drug candidate, the cost associated with the drug discovery process, and the probability of success in clinical trials. However, [1] does not explicitly state which evaluation metrics are used.

One of the key challenges in evaluating LLMs for drug research is the lack of standardized datasets and benchmarks. While some publicly available datasets exist, they often do not fully capture the complexity and diversity of real-world drug discovery problems. This makes it difficult to compare the performance of different LLMs and to assess their generalizability to new tasks and datasets. Furthermore, evaluation metrics may not always align with the ultimate goals of drug discovery, such as identifying safe and effective drugs that can improve patient outcomes. For example, an LLM that achieves high accuracy in predicting drug-target interactions may not necessarily lead to the discovery of a novel drug with therapeutic potential.

Despite these challenges, LLMs hold great promise for transforming the drug discovery process. As LLMs become more powerful and sophisticated, they are likely to play an increasingly important role in all stages of drug development, from target identification to clinical trials. However, it is important to carefully consider the limitations and potential biases of LLMs and to develop appropriate evaluation metrics to ensure that they are used responsibly and effectively. Further research is needed to develop standardized datasets and benchmarks, to improve the interpretability and explainability of LLMs, and to address ethical considerations such as data privacy and bias in training data. In addition, future research should focus on integrating LLMs with other AI techniques, such as deep learning and reinforcement learning, to create more powerful and versatile drug discovery platforms. The lack of details in the reviewed articles about the specific datasets and tools used by

PharmAgents makes it difficult to fully assess its potential impact and limitations, highlighting a need for more transparency and reproducibility in future research.

Resources and Datasets

Resources and datasets are fundamental to the advancement of LLM applications in drug research. The reviewed articles highlight the reliance on a diverse range of resources, from specialized databases to computational tools and hardware infrastructure, that enable the training, evaluation, and deployment of LLMs in pharmaceutical contexts. A key resource identified is domain-specific datasets, essential for training LLMs to understand and reason about complex biological and chemical information [1, 2]. For instance, in the PharmAgents framework [1], the target discovery module utilizes both a Drug Target Database and the Unified Protein Database (UniProt) to identify potential therapeutic targets. These databases provide crucial information about protein sequences, functions, and interactions, enabling the LLM-based disease expert to identify relevant targets. The lead identification stage within PharmAgents necessitates access to chemical compound databases, such as ZINC, to identify potential drug candidates [1]. These databases contain information on millions of commercially available and synthetically tractable compounds, which are essential for training LLMs to generate and evaluate novel drug candidates.

Moreover, specialized LLMs are trained on scientific languages such as SMILES (Simplified Molecular Input Line Entry System) and FASTA (representing nucleotide or protein sequences), demonstrating the need for data encoded in formats interpretable by both humans and machines [2]. These specialized languages allow LLMs to directly process and reason about molecular structures and biological sequences, leading to more accurate predictions of drug efficacy and safety. The PharmAgents framework further exemplifies the need for specialized computational tools, including molecular docking software and simulation packages [1]. These tools enable the virtual screening and evaluation of drug candidates, providing crucial insights into their binding affinity, ADMET (absorption, distribution, metabolism, excretion, and toxicity) properties, and overall therapeutic potential. The integration of these tools into the LLM-driven workflow allows for a more comprehensive and efficient drug discovery process.

Furthermore, the review in [2] implicitly suggests the importance of access to high-performance computing (HPC) infrastructure, especially when dealing with large-scale datasets and complex models, although this aspect is not explicitly detailed. Training and deploying LLMs for drug discovery often require significant computational resources, including GPUs (Graphics Processing Units) and TPUs (Tensor Processing Units), to handle the vast amounts of data and perform computationally intensive tasks such as model training and inference. In addition to datasets and computational resources, the availability of pre-trained models and open-source libraries is also crucial for facilitating the adoption of LLMs in drug research. These resources allow researchers to leverage existing knowledge and accelerate the development of new applications. The size and quality of these datasets influence the performance and generalizability of the resulting LLMs [2]. Biases present in the training data can also lead to biased predictions and unfair outcomes, highlighting the importance of carefully curating and evaluating datasets. The limitations of available resources, such as the lack of comprehensive datasets for certain diseases or the computational cost of training large models, can also pose significant challenges for researchers in this field. Overcoming these challenges will require collaborative efforts to create and share high-quality datasets, develop more efficient algorithms, and provide access to affordable computing infrastructure.

Challenges and Limitations

Despite the transformative potential of LLMs in drug research, several challenges and limitations must be addressed to ensure their responsible and effective implementation. These challenges span data-related issues, methodological shortcomings, computational constraints, and ethical considerations. Data-related limitations

represent a significant hurdle, as the performance of LLMs is heavily dependent on the quality and quantity of training data [2]. The availability of comprehensive, high-quality, and well-annotated datasets in specific areas of drug research, such as target identification or drug-target interaction prediction, remains a concern. Furthermore, biases present in the training data can be amplified by LLMs, leading to skewed predictions and potentially hindering drug discovery efforts [2]. The risk of overfitting to specific datasets is also a valid concern. PharmAgents, while promising, could potentially suffer from biases present in the databases it queries, impacting the target discovery and lead optimization stages [1].

Methodological limitations also pose challenges to the widespread adoption of LLMs in drug research. The complexity of biological systems and the inherent uncertainty in drug discovery make it difficult to fully capture the underlying mechanisms and relationships using LLMs [2]. The "black box" nature of some LLMs can also hinder interpretability and trust, making it challenging to understand the reasoning behind their predictions. This lack of transparency can be particularly problematic in critical decision-making stages, such as lead optimization and preclinical evaluation. PharmAgents addresses this partially by using explainable agents, but the extent to which these explanations are truly comprehensive and reliable needs further scrutiny [1]. Validating the predictions and recommendations generated by LLMs requires rigorous experimental validation, which can be costly and time-consuming.

Computational constraints represent another set of challenges. Training and deploying large LLMs requires significant computational resources, including powerful hardware and specialized software [2]. This can limit accessibility for researchers with limited resources and hinder the scalability of LLM-based drug discovery platforms. The computational cost of simulating the entire drug discovery pipeline, as envisioned by PharmAgents [1], could be substantial, requiring optimized algorithms and efficient hardware infrastructure.

Ethical considerations are paramount in the application of LLMs to drug research. Data privacy is a major concern, as LLMs may be trained on sensitive patient data or proprietary information [2]. Ensuring the security and confidentiality of such data is crucial. The potential for misuse of LLMs, such as generating misleading or harmful information, also needs to be addressed. Furthermore, the ethical implications of automating drug discovery processes and potentially displacing human researchers need careful consideration. The use of LLMs to accelerate drug discovery raises questions about equitable access to new treatments and the potential for exacerbating existing health disparities [2].

Addressing these challenges requires a multi-faceted approach. Mitigation strategies include developing robust data governance frameworks, incorporating fairness and bias detection techniques into LLM training pipelines, and promoting transparency and explainability in LLM-based systems. PharmAgents attempts to tackle some of these by providing domain-specific knowledge and self-evolution capabilities [1], but it is unclear to what extent these features fully address the ethical and data-related concerns. Developing standardized evaluation metrics and benchmarking datasets is also crucial for objectively assessing the performance of LLMs in drug research. Ultimately, the responsible and ethical implementation of LLMs in drug research requires collaboration between researchers, policymakers, and industry stakeholders to ensure that these powerful tools are used to benefit society as a whole. A recurring obstacle across the papers is the lack of comprehensive validation studies to confirm the in-silico predictions made by LLMs. This significantly limits the translational potential of these models. The implications of these limitations are that, without careful consideration and proactive mitigation, the promise of LLMs in accelerating drug discovery may be undermined by biased results, lack of transparency, and ethical concerns.

Proposed Research Pipeline

The proposed research pipeline aims to leverage the power of LLMs to revolutionize drug discovery, addressing the challenges of high costs, long timelines, and low success rates that plague traditional pharmaceutical research. This pipeline builds upon the concept of autonomous agents as demonstrated in [1], but also incorporates a more structured approach to data integration and validation, inspired by the comprehensive

review of LLM applications in [2]. By systematically guiding the LLM through a series of carefully designed stages, this pipeline seeks to enhance the efficiency, accuracy, and explainability of the drug discovery process. The pipeline aims to mitigate limitations of existing models, such as data scarcity and the need for robust validation methods highlighted in [2] and [1].

Stage 1: Data Curation and Integration. This stage focuses on compiling and preprocessing diverse datasets relevant to drug discovery. Drawing from the insights in [1], this includes drug target databases, protein databases (e.g., UniProt), chemical compound databases (e.g., ChEMBL, PubChem), and scientific literature databases (e.g., PubMed). The data will be standardized and integrated into a unified knowledge graph. This ensures that the LLM has access to a comprehensive and structured representation of relevant information. The curation process will involve automated data cleaning, entity linking, and relationship extraction. Validation at this stage will involve assessing the completeness and accuracy of the integrated knowledge graph using established benchmarks and expert review.

Stage 2: Target Identification and Validation. Inspired by the target discovery module in PharmAgents [1], this stage uses LLMs to identify potential drug targets based on disease descriptions. The LLM will be prompted with detailed information about a disease, including its symptoms, genetic factors, and molecular pathways. The LLM will then leverage the curated knowledge graph from Stage 1 to identify potential therapeutic targets. This stage will involve a RAG (Retrieval-Augmented Generation) approach, where the LLM retrieves relevant information from the knowledge graph to generate hypotheses about potential drug targets. Identified targets will be ranked based on their relevance to the disease and their druggability. Evaluation will involve comparing the LLM-generated targets with known drug targets for the disease and assessing their potential for therapeutic intervention using metrics like novelty and biological plausibility.

Stage 3: Lead Compound Generation and Optimization. This stage focuses on generating and optimizing lead compounds that can interact with the identified drug targets. As highlighted in [2], LLMs can be used to design novel molecules with desired properties. This stage will employ generative models, such as those trained on SMILES strings, to generate novel chemical structures. The generated compounds will then be evaluated using computational methods, such as molecular docking and molecular dynamics simulations, to assess their binding affinity and selectivity for the target protein. The LLM will iteratively optimize the chemical structures based on the results of these simulations. The optimization process can utilize techniques similar to those in PharmAgents [1], such as reinforcement learning, to guide the LLM towards compounds with improved properties. Intermediate validation points will be quantitative structure-activity relationship (QSAR) modeling to predict activity based on generated structures.

Stage 4: Preclinical Evaluation and Prediction. Building upon the preclinical evaluation module in [1], this stage uses LLMs to predict the efficacy and safety of the lead compounds in preclinical models. The LLM will be trained on preclinical data, including in vitro and in vivo studies, to predict the effects of the compounds on various biological endpoints. This stage will involve natural language processing (NLP) techniques to extract relevant information from preclinical reports. The LLM will then use this information to predict the efficacy and safety of the compounds. The predictions will be validated using external datasets and experimental data, when available. Evaluation will involve comparing the LLM predictions with experimental results and assessing the accuracy and reliability of the predictions.

Stage 5: Clinical Trial Design and Optimization. As outlined in [2], LLMs can be used to optimize clinical trial design and improve patient selection. This stage will use LLMs to analyze clinical trial data and identify factors that influence the efficacy and safety of the drug. The LLM will also be used to optimize the clinical trial protocol, including the patient population, dosage, and duration of treatment. The LLM will leverage information from previous clinical trials and real-world data to predict the outcomes of the trial. The optimized trial designs will then be assessed by human experts. Evaluation involves simulating clinical trials using LLMs and comparing outcomes to published results for similar drugs and patient populations.

Stage 6: Explainability and Interpretability Analysis. This stage focuses on enhancing the explainability and interpretability of the LLM's predictions. Given the importance of transparency highlighted in [1], techniques

such as attention mechanisms and layer-wise relevance propagation will be used to identify the key factors that influenced the LLM's decisions. The LLM's reasoning will be documented and presented in a clear and concise manner. The explanations will be validated by domain experts to ensure their accuracy and relevance. This will help to build trust in the LLM's predictions and facilitate the translation of research findings into clinical practice. This stage could also incorporate a feedback loop, using expert feedback on the LLM's explanations to improve its reasoning and decision-making processes.

A key challenge for this proposed pipeline is the integration of heterogeneous data sources and formats, which requires significant effort in data curation and standardization, as highlighted in [2]. Mitigation strategies include developing robust data cleaning and transformation pipelines, as well as using standardized ontologies and vocabularies to ensure consistency and interoperability. Another challenge is the computational cost of training and deploying large LLMs, as well as the potential for bias in the training data. To address these challenges, we will employ techniques such as transfer learning and fine-tuning to reduce the training time and resource requirements. We will also carefully evaluate the LLM for bias and implement strategies to mitigate any identified biases. Furthermore, rigorous validation of each stage of the pipeline is crucial to ensure the reliability and accuracy of the results. Incorporating human experts in the loop for critical decision points will also add a layer of validation and control.

Implementing this proposed pipeline holds significant promise for accelerating drug discovery and reducing the cost and time associated with traditional pharmaceutical research. By leveraging the power of LLMs, this pipeline can identify novel drug targets, generate and optimize lead compounds, predict preclinical and clinical outcomes, and provide explanations for its predictions. The successful implementation of this pipeline could lead to the development of new and more effective treatments for a wide range of diseases.

Conclusion

In conclusion, this review has explored the burgeoning field of LLMs in drug research, synthesizing key themes, dominant methods, and major challenges based on the analyzed literature. The reviewed papers highlight the potential of LLMs to revolutionize traditional pharmaceutical methodologies by automating drug discovery workflows [1], enhancing target-disease linkage, interpreting biomedical data, improving drug molecule design, predicting drug efficacy and safety, and streamlining clinical trials [2]. A dominant method identified is the use of multi-agent systems driven by LLMs, exemplified by PharmAgents [1], which simulates the entire pharmaceutical research process within a virtual environment. However, significant challenges remain, including data-related limitations, methodological shortcomings, computational constraints, and ethical considerations [2].

Overall, the state of research on LLMs in drug research, as reflected in the reviewed articles, is promising but still in its early stages. While LLMs have demonstrated potential in various aspects of drug discovery and development, their widespread adoption is hindered by several factors. The availability of comprehensive and high-quality training data is a major bottleneck [2], as is the need for more robust evaluation metrics to assess the performance of LLMs in drug research tasks. Furthermore, the lack of transparency and interpretability of some LLM models raises concerns about their reliability and trustworthiness. The PharmAgents framework [1] represents a significant step towards autonomous and explainable pharmaceutical research, but further research is needed to validate its capabilities and address its limitations.

Effective research approaches in this area emphasize the importance of integrating domain-specific knowledge with advanced LLM techniques [1]. The development of specialized LLMs trained on specific scientific languages and datasets [2] is crucial for achieving high performance in drug research tasks. Furthermore, the proposed research pipeline underscores the need for a structured approach to data integration and validation, ensuring the reliability and accuracy of LLM-generated predictions. The pipeline also highlights the importance of incorporating experimental validation and feedback loops to refine LLM models and improve their performance over time.

Promising directions for future research include exploring novel architectures and training strategies for LLMs, developing more sophisticated evaluation metrics tailored to drug research tasks, and addressing the ethical considerations surrounding the use of LLMs in drug discovery. Further research is needed to improve the interpretability and explainability of LLM models, enabling researchers to understand the reasoning behind their predictions and identify potential biases. Another promising direction is the development of more robust and scalable multi-agent systems, such as PharmAgents, that can seamlessly integrate with existing drug discovery workflows [1]. Finally, there is a need for more comprehensive benchmarking studies to compare the performance of different LLM models and identify best practices for their application in drug research. Expanding the scope of datasets used to train and test these models, and ensuring these datasets are representative and unbiased, is critical. The advancement of generative AI techniques to design novel drug candidates with desired properties is another promising avenue.

Advancing research in LLMs for drug discovery holds immense potential. By accelerating the drug discovery pipeline, reducing costs, and improving success rates, LLMs could transform the pharmaceutical industry and ultimately improve patient outcomes. The ability to rapidly identify promising drug candidates, predict their efficacy and safety, and optimize clinical trial designs could lead to the development of new treatments for a wide range of diseases [2, 1]. Moreover, the use of LLMs could democratize drug discovery, enabling smaller research teams and academic institutions to participate in the development of new therapies. Overcoming the current limitations and challenges will pave the way for a future where AI-driven drug discovery is a reality, leading to faster, more efficient, and more effective treatments for diseases that affect millions worldwide.

References

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